

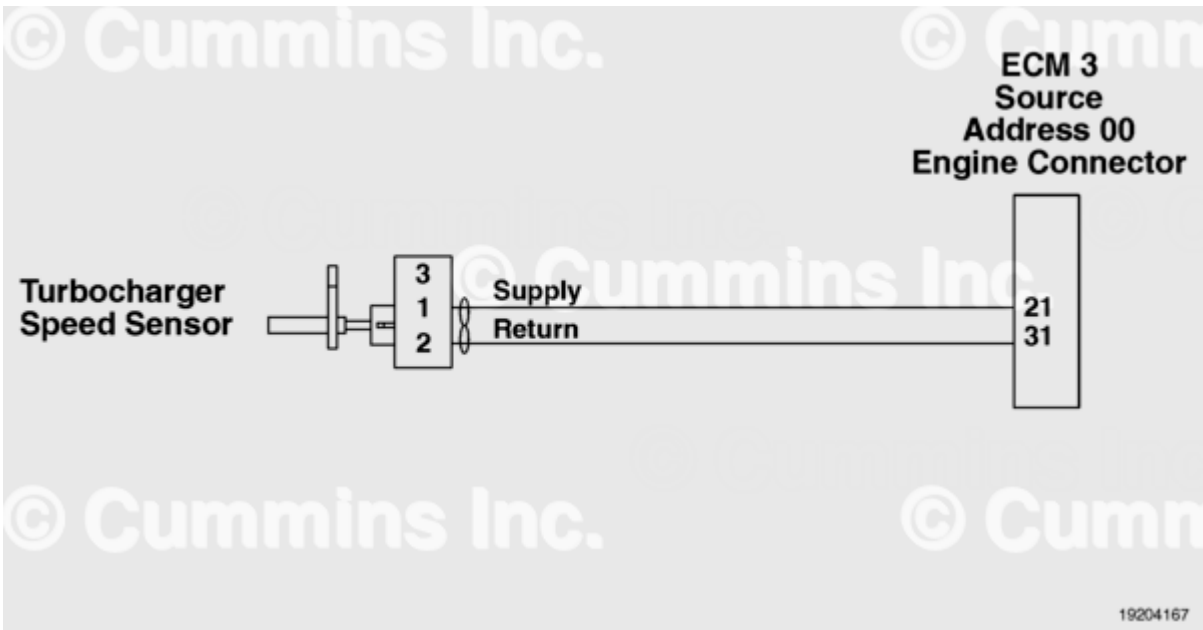
Fault Code: 595

Turbocharger 1 Speed - Data Valid But Above Normal Operating Range - Moderately Severe Level

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 595 PID(P): P103 SPN: 103 FMI: 0/16 Lamp: Amber SRT:	Turbocharger 1 Speed - Data Valid But Above Normal Operating Range - Moderately Severe Level. High turbocharger speed has been detected by the engine control module (ECM).	Possible reduced engine performance.



Turbocharger Speed Sensor Circuit

Circuit Description

The turbocharger speed sensor is a variable reluctance speed sensor. It consists of a coil of wire and an iron core. The target on the turbocharger shaft is a ground flat in the center of the shaft. As the flat on the turbocharger shaft spins past the speed sensor, a signal is generated. The ECM interprets this signal and converts it to a turbocharger speed reading.

Component Location

The turbocharger speed sensor is mounted in the center housing of the turbocharger.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is operating.

Conditions For Setting The Fault Codes

The ECM detected the turbocharger speed was greater than 150,000 rpm for more than 5 seconds.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE lamp and/or the malfunction indicator lamp (MIL) immediately when the diagnostic runs and fails.
- The ECM will estimate turbocharger speed.

Conditions For Clearing The Fault Code

To validate the repair, start the engine and use the accelerator pedal to accelerate the engine speed to high idle (100 percent). Hold the engine speed at high idle for at least 20 seconds.

- The fault code status displayed by INSITE™ electronic service tool will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp after the diagnostic runs and passes.
- For On-Board Diagnostics (OBD) engines, the ECM will extinguish the MIL after three consecutive trips where the diagnostic runs and passes.
- The "Reset All Faults" command in INSITE™ electronic service tool can be used to clear active and inactive faults, as well as extinguish the MIL for OBD applications.

Shoptalk

Possible causes of this fault code include:

- Damaged turbocharger speed sensor
- Damaged engine harness
- Damaged ECM.

If the fault occurs intermittently, look for causes of intermittent open or short circuits in the turbocharger speed sensor circuit (including the speed sensor pigtail connector).

Refer to Troubleshooting Fault Code t05-595. (/qs3/pubsys2/xml/en/procedures/122/122-t05-595.html)

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